

Dobrushin's Wilson criterion: an explicit strong-coupling gap at $g^2 > 444$ and an open block test

The finite-volume criteria of the finite-verification note have a single-site ancestor, Dobrushin's uniqueness condition (Theory Probab. Appl. 13 (1968) 197), and for the Wilson action it can be checked by hand. The comparison below uses four-dimensional lattices and spatial tori with sides at least four, as in the polymer note. The conditional distribution of one link given all the others is

$$d\mu_\ell(U \mid \omega) \propto \exp \left[\frac{\beta_W}{N} \operatorname{Re} \operatorname{tr} (U M_\ell(\omega)) \right] d\mu(U), \quad M_\ell = \sum_{p \ni \ell} S_p,$$

the sum of the six staples of ℓ in four dimensions. Changing one neighbouring link changes one staple, and the resulting total-variation distance between the two conditional distributions is at most $e^{4\beta_W} - 1$, independently of N . Each link has 18 neighbours that share a plaquette with it, so Dobrushin's coefficient is

$$\alpha = \sup_{\ell} \sum_{\ell'} \rho_{\ell\ell'} \leq 18(e^{4\beta_W} - 1),$$

and the condition $\alpha < 1$ holds for $\beta_W < \frac{1}{4} \log \frac{19}{18} \simeq 0.01352$, that is

$$g^2 > \frac{8N}{\log(19/18)}; \quad g^2 > 148N = 444 \quad (SU(3)) \text{ is a sufficient rounded bound.}$$

Dobrushin's theorem gives a unique Gibbs state, Künsch's theorem (Commun. Math. Phys. 84 (1982) 207) gives exponential decay of all truncated correlations at rate $\log(1/\alpha)$ per lattice unit, and link reflection positivity turns the rate into a gap:

$$\Delta_W \geq \frac{\hbar c}{a} \log \frac{1}{18(e^{24/g^2} - 1)} \simeq \frac{\hbar c}{a} \log \frac{g^2}{432} \quad (g^2 > 444),$$

uniformly in the volume. This is a better threshold than the polymer expansion's sufficient integer bound 1059 (1056 is linearized). Its rate grows like $\log(g^2/432)$, whereas the polymer rate grows like $4\log(g^2/1056)$ at large g^2 . The block version of the same condition, the Dobrushin–Shlosman constructive criterion (in *Statistical Physics and Dynamical Systems*, Birkhäuser 1985, 347), replaces the single link by a box V of links and is expected to hold deeper into the intermediate region as V grows; it is the quantity a finite verification would compute, and Section 4 states it as a specification. All references at metadata level; constants explicit; nothing promoted.

Correction, 2026-10-02 (comparison rates). The earlier table printed the leading logarithms as rigorous lower bounds. The exact exponential-activity rates are given in the corrected table in §3; the logarithms are asymptotic estimates. This also distinguishes the exact single-link condition from its rounded sufficient threshold.

1. The one-link conditional distribution

Fix all links except ℓ . The plaquettes containing ℓ are six in four dimensions, and each contributes $\operatorname{Re} \operatorname{tr}(U_\ell S_p)$ with $S_p \in SU(N)$ the product of the other three links of p in the appropriate order. Hence the conditional density is $e^{f_\omega(U)}$ with $f_\omega(U) = (\beta_W/N) \operatorname{Re} \operatorname{tr}(U M_\ell)$, $M_\ell = \sum_{p \ni \ell} S_p$, normalized by $Z(\omega) = \int e^{f_\omega} d\mu$. Gauge invariance of the interaction is visible in the fact that a gauge transformation at an endpoint of ℓ rotates M_ℓ and the Haar measure absorbs it; nothing more is needed.

2. The Dobrushin coefficient

Lemma 1. For probability densities $p \propto e^f$, $q \propto e^{f'}$ with respect to the same measure, $\|p - q\|_{\text{TV}} \leq e^{\text{osc}(f-f')} - 1$.

Proof. $p/q = e^{f-f'} Z_{f'}/Z_f \leq e^{\sup(f-f')} e^{\sup(f'-f)} = e^{\text{osc}(f-f')}$, so $p - q \leq (e^{\text{osc}} - 1)q$ and $\|p - q\|_{\text{TV}} = \int (p - q)_+ \leq e^{\text{osc}} - 1$. $\square$

Lemma 2. If ω, ω' differ only at a link ℓ' sharing a plaquette with ℓ , then $\text{osc}_U(f_\omega - f_{\omega'}) \leq 4\beta_W$.

Proof. One staple changes, $M_\ell - M'_\ell = S_p - S'_p$ with $S_p, S'_p \in SU(N)$, so $|f_\omega(U) - f_{\omega'}(U)| = (\beta_W/N) |\text{Re tr}(U(S_p - S'_p))| \leq (\beta_W/N) \|S_p - S'_p\|_1 \leq (\beta_W/N) \cdot 2N$, the trace norm of a difference of two unitaries being at most $2N$; the oscillation is at most twice the supremum. $\square$

If ℓ' shares no plaquette with ℓ , the conditional distribution at ℓ does not depend on it. Each link lies in six plaquettes with three further links each, and the six plaquettes share no link besides ℓ , so there are exactly 18 links ℓ' with $\rho_{\ell\ell'} \neq 0$. Hence

$$\alpha = \sup_\ell \sum_{\ell'} \rho_{\ell\ell'} \leq 18(e^{4\beta_W} - 1),$$

and the upper bound on α is strictly below one precisely when $\beta_W < \frac{1}{4} \log(19/18)$. The reported 0.01352 and $148N$ are rounded figures; $g^2 > 148N$ is sufficient since $\log(19/18) > 2/37$.

3. From the condition to the gap

Dobrushin's theorem: $\alpha < 1$ implies a unique Gibbs state on $\mathbb{Z}^4$ and on every torus. The covariance estimate of Föllmer (J. Funct. Anal. 46 (1982) 387; Künsch, loc. cit., for the decay statements): under $\alpha < 1$, for local observables F, G ,

$$|\langle F; G \rangle| \leq \sum_{\ell \in \text{supp } F, \ell' \in \text{supp } G} \delta_\ell F [(1 - \rho)^{-1}]_{\ell\ell'} \delta_{\ell'} G,$$

with $\delta_\ell F$ the oscillation of F in the link ℓ , and since $\rho_{\ell\ell'}^n$ vanishes unless the link distance is at most n lattice units, $[(1 - \rho)^{-1}]_{\ell\ell'} \leq \alpha^{d(\ell, \ell')}/(1 - \alpha)$. Truncated correlations at time separation T therefore decay at least like $\alpha^{T/a}$, rate $m = \log(1/\alpha)/a$. The Wilson transfer matrix is positive by link reflection positivity, and for a local A orthogonal to the vacuum $\langle A, \mathcal{T}^n A \rangle \leq C_A \alpha^n$ forces its spectral measure to vanish above α , so on the cyclic subspace of local observables

$$\Delta_W = -\frac{\hbar c}{a} \log \|\mathcal{T}|_{\Omega^\perp}\| \geq \frac{\hbar c}{a} \log \frac{1}{\alpha} \geq \frac{\hbar c}{a} \log \frac{1}{18(e^{24/g^2} - 1)} \quad (SU(3)).$$

route	rigorous threshold ($SU(3)$)	gap bound
polymer expansion, all representations	$g^2 \geq 1059$ (sufficient integer bound)	$(\hbar c/a) 4 \log[1/(176(e^{6/g^2} - 1))]$
Dobrushin single-site condition	$g^2 > 24/\log(19/18)$; $g^2 > 444$ suffices	$(\hbar c/a) \log[1/(18(e^{24/g^2} - 1))]$

The two are complementary: the Dobrushin route enters earlier, the polymer route's rate is larger at sufficiently large coupling, rather than throughout their overlap. For $U(1)$ the same computation gives $g^2 > 148$ with $N = 1$.

4. The block criterion, as a specification

Corrected. The single-link form below is emptied by Elitzur's theorem: the marginal of any link with an interior endpoint is Haar for every boundary condition,

so $\rho_V(x, y) = 0$ there and the sum sees only the boundary layer. The form to verify is the mixing condition on sub-blocks, organized around the conditional distribution of small Wilson loops, and the boxes needed are of side 3 to 5 at $\beta_W \simeq 5.7$; see the bands note.

The Dobrushin–Shlosman constructive criterion replaces the single link by a finite box V of links. Writing $\mu_V(\cdot|\omega)$ for the Gibbs measure in V with boundary condition ω and, for $x \in V$ and $y \notin V$,

$$\rho_V(x, y) = \sup_{\omega, \omega' \text{ differ only at } y} \|\mu_V(\cdot|\omega)|_x - \mu_V(\cdot|\omega')|_x\|_{TV},$$

the criterion is that for some finite V

$$\sum_{x \in V} \sum_{y \notin V} \rho_V(x, y) < |V|,$$

up to the normalization convention of the original, and it implies the same conclusions as the single-site condition, which is the case $V = \{\ell\}$. Its content is that the total influence of the boundary on the box, summed over the box, is less than the size of the box; in a mixing phase this holds once the side of V exceeds a few correlation lengths, which is why it can hold at couplings where the single-site condition fails. Dobrushin and Shlosman’s complete analyticity (*Statistical Physics and Dynamical Systems*, Birkhäuser 1985, 371; J. Stat. Phys. 46 (1987) 983) is the statement that some such V exists, together with its equivalent forms.

What a verification computes. For a box V of side R (about $3R^4$ links) and a coupling β_W : for each boundary link y and each interior link x , a rigorous upper bound on $\rho_V(x, y)$, which is a total-variation distance between two marginals of Gibbs measures on $SU(3)^{|V|}$; then the double sum against $|V|$. Lemma 1 bounds each $\rho_V(x, y)$ by $e^{\text{osc}} - 1$ of the log-density difference of the two marginals, but for $|V| > 1$ that oscillation involves the integral over the other links of V and is what has to be bounded rigorously. The openness radius in the interaction norm $\sup_{\ell} \sum_{p \ni \ell} \|\Phi_p\|_{\infty}$ follows from the margin $|V| - \sum \rho_V$ by the same lemma.

Where it would have to hold. The numerical evidence places the $SU(3)$ crossover at $\beta_W \simeq 5\text{--}6.5$ with ξ/a of order 1 to 10; a verification at those couplings would need R of order 10–30, that is $10^4\text{--}10^6$ links, and rigorous bounds on the corresponding integrals. That is far beyond present practice, and this repository performs no numerics. The specification is nonetheless complete: the quantity, the boxes, the inequality, and the two lemmas that convert its margin into a gap and into an openness radius.

5. Consequence for STATE

The strong-coupling side now has a third, elementary proof with the best rigorous threshold for the Wilson transfer matrix, $g^2 > 444$ for $SU(3)$, and the finite verification of the intermediate region has a precise statement: the Dobrushin–Shlosman block criterion for the Wilson interaction, whose single-link case is proved here. The two numbers of the finite-verification note are unchanged, and the second of them, g_{DS}^2 , is exactly the smallest coupling at which the block criterion has been verified for some V . The October 2 comparison uses the exact rates in §3; their leading logarithms remain estimates. All of these are fixed-cutoff Wilson transfer statements, distinct from Kogut–Susskind and a continuum gap.